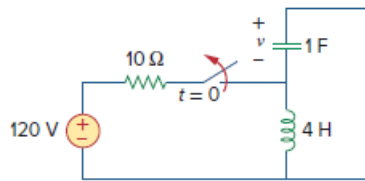


Problem

Obtain $v(t)$ for $t > 0$ in the circuit of Fig. 16.46.

Figure 16.46:



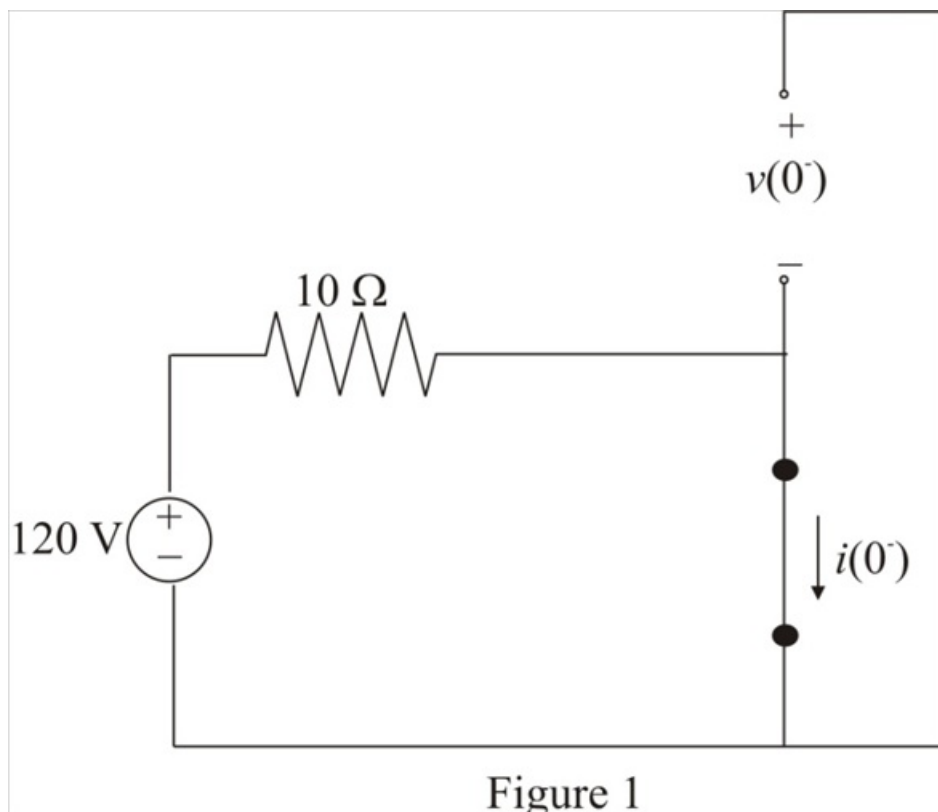
Step-by-step solution

Step 1 of 8

Refer to Figure 16.46 in the text book.

Draw the circuit diagram at $t < 0$.

At $t < 0$, inductor acts as short circuit and capacitor acts as open circuit.



Step 2 of 8

Notice that the value of the voltage $v(0^-)$ is 0 V since its terminals are shorted.

Calculate the value of the current $i(0^-)$.

$$i(0^-) = \frac{120}{10} \\ = 12 \text{ A}$$

Therefore, the value of the current $i(0^-)$ is 12 A.

Draw the circuit diagram at $t > 0$. Represent the circuit parameters in Laplace transform.

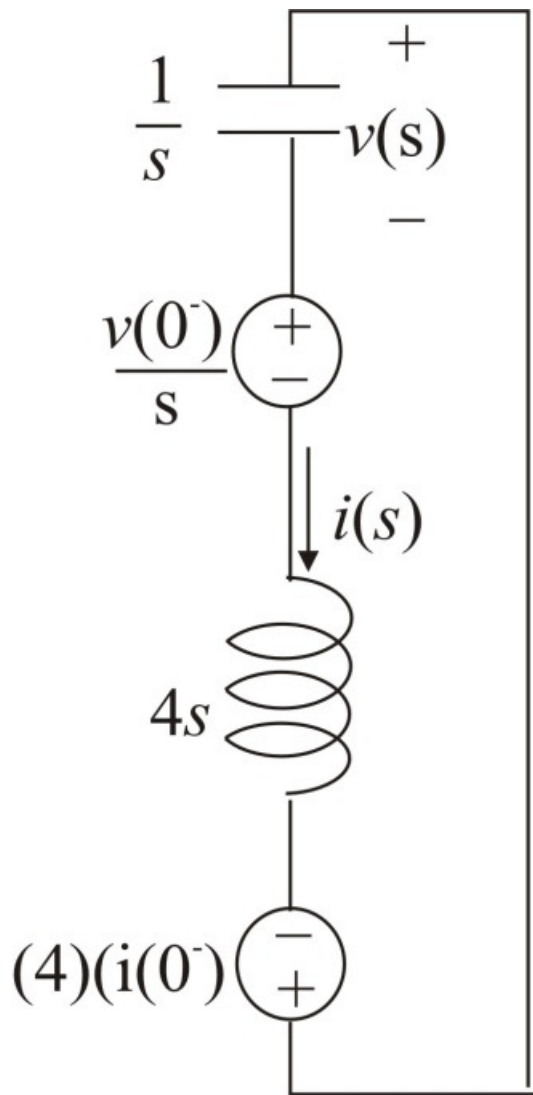


Figure 2

Apply Kirchhoff's Voltage Law in the loop.

$$i(s) \left(4s + \frac{1}{s} \right) - (4)(i(0^-)) + \frac{v(0^-)}{s} = 0$$

Substitute 12 A for $i(0^-)$ and 0 V for $v(0^-)$.

$$i(s) \left(4s + \frac{1}{s} \right) - (4)(12) + \frac{0}{s} = 0$$

$$i(s) \left(\frac{4s^2 + 1}{s} \right) - 48 = 0$$

$$i(s)(4s^2 + 1) = 48s$$

$$i(s) = \frac{48s}{4s^2 + 1}$$

Therefore, the expression for the current $i(s)$ is $\frac{48s}{4s^2 + 1}$.

Calculate the value of the voltage $v(s)$.

$$v(s) = i(s) \left(\frac{1}{s} \right)$$

Substitute $\frac{48s}{4s^2 + 1}$ for $i(s)$.

$$\begin{aligned} v(s) &= \left(\frac{48s}{4s^2 + 1} \right) \left(\frac{1}{s} \right) \\ &= \frac{48}{4s^2 + 1} \\ &= \frac{48}{4(s^2 + 0.25)} \end{aligned}$$

$$v(s) = \frac{12}{(s + j0.5)(s - j0.5)} \dots\dots (1)$$

Step 6 of 8

Consider $\frac{12}{(s + j0.5)(s - j0.5)}$ to be $\frac{A_1}{s + j0.5} + \frac{A_2}{s - j0.5}$.

$$\frac{10}{(s + j0.5)(s - j0.5)} = \frac{A_1}{s + j0.5} + \frac{A_2}{s - j0.5}$$

Apply residue method to find the value of the constant A_1 .

$$\begin{aligned} A_1 &= (s + j0.5) \left(\frac{12}{(s + j0.5)(s - j0.5)} \right) \Bigg|_{s = -j0.5} \\ &= \frac{12}{(s - j0.5)} \Bigg|_{s = -j0.5} \\ &= \frac{12}{-j0.5 - j0.5} \\ &= \frac{12}{-j1} \end{aligned}$$

$$A_1 = j10$$

Therefore, the value of the constant A_1 is $j12$.

Step 7 of 8

Apply residue method to find the value of the constant A_2 .

$$\begin{aligned} A_2 &= (s - j0.5) \left(\frac{12}{(s + j0.5)(s - j0.5)} \right) \Bigg|_{s = j0.5} \\ &= \frac{12}{(s + j0.5)} \Bigg|_{s = j0.5} \\ &= \frac{12}{j0.5 + j0.5} \\ &= \frac{12}{j1} \end{aligned}$$

$$A_2 = -j10$$

Therefore, the value of the constant A_2 is $-j12$.

Step 8 of 8

Recall equation (1).

$$v(s) = \frac{12}{(s + j0.5)(s - j0.5)}$$

Substitute $\frac{A_1}{s + j0.5} + \frac{A_2}{s - j0.5}$ for $\frac{12}{(s + j0.5)(s - j0.5)}$.

$$v(s) = \frac{A_1}{s + j0.5} + \frac{A_2}{s - j0.5}$$

Substitute $-j12$ for A_2 and $j12$ for A_1 .

$$v(s) = \frac{-j12}{s + j0.5} + \frac{j12}{s - j0.5}$$

Apply inverse Laplace transform.

$$\begin{aligned} v(t) &= (j12)(e^{j0.5t} - e^{-j0.5t})u(t) \\ &= (-24)\left(\frac{e^{j0.5t} - e^{-j0.5t}}{2j}\right)u(t) \\ &= (-24)\sin(0.5t)u(t) \\ &= 24\cos(0.5t + 90^\circ)u(t) \text{ V} \end{aligned}$$

Therefore, the value of the voltage $v(t)$ is $\boxed{24\cos(0.5t + 90^\circ)u(t) \text{ V}}$.